

Experimental execution summary

Tasnimul Hasan

Completed single-seed, 200-epoch training runs. Model: standard MK-UNet imported from the pinned official code. Hardware: NVIDIA GeForce RTX 5070 Laptop GPU, 8 GB VRAM. Runtime: Python 3.13, PyTorch 2.11.0+cu128.

Experiment	Test n	Dice / IoU	Best epoch
ClinicDB	62	91.28% / 85.19%	200
CWFID	21	87.98% / 78.59%	128

Reference run: 70.2 minutes. Agricultural run: 6.8 minutes. Times include training, validation, checkpoint writes, and final evaluation; they are not pure inference timings.

The ClinicDB run achieved 91.28% Dice and 85.19% IoU. Relative to the published 93.48% Dice reference [1], the difference is -2.20 percentage points. This comparison is descriptive: the publication averages five runs, whereas this study uses one seed and corrects evaluation and resizing behavior in the current public code.

With a fixed sigmoid threshold of 0.5, test Dice is 91.28% on ClinicDB and 87.98% on CWFID. The primary scores instead use the upstream per-image min-max normalization followed by 0.5 thresholding. This extra result exposes sensitivity to output scaling.

Setup and execution

Run scripts/download_clinicdb.py, then scripts/prepare_data.py all. Train with python -m screening.run train --config configs/clinicdb_reference.json and the corresponding cwfid_domain.json configuration. Use --resume to continue the same configuration from last.pt. See README.md for the exact environment and full workflow.

Evidence index

Each run directory contains provenance.json, history.jsonl, best.pt, last.pt, metrics.json, test/per_image.csv, and test/predictions/. Dataset manifests record split assignments and file hashes. sources/protocol_audit.md documents deviations, dataset corrections, and interpretation limits. Tests cover metric arithmetic, gradient direction, mask foreground, split overlap, and native-resolution evaluation.

[Code repository](#) | [Reproduction instructions](#).

Reproducibility status

The experimental summary, three-page technical report, and six-page proposal are linked to the published configurations, manifests, run histories, per-image measurements, and verification workflow. The repository link appears above.